

2 (a) Define the Young modulus of a material.

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..... [1]

- (b) A long, thin thread of high-density polythene has a uniform cross-section. The thread is suspended from one end so that it hangs vertically.

The thread has a mass of 0.018 kg and a cross-sectional area of $2.6 \times 10^{-6} \text{ m}^2$.
High-density polythene has a density of 960 kg m^{-3} and a Young modulus of $8.0 \times 10^8 \text{ Pa}$.

- (i) Show that the length of the thread is 7.2 m.

[2]

- (ii) Show that the stress at the top of the thread due to its own weight is $6.8 \times 10^4 \text{ Pa}$.

[2]

- (iii) Figure 2.1 shows the variation with distance from the top of the thread of the stress in the thread due to its own weight.

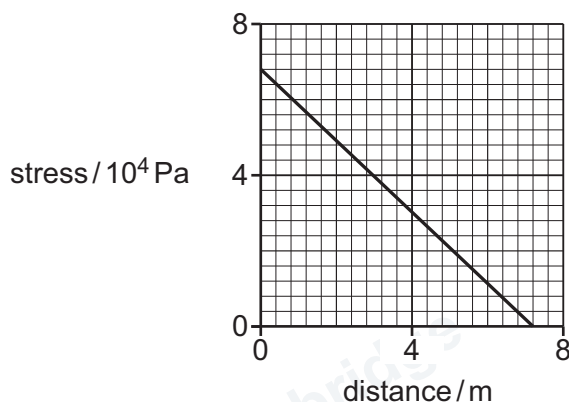


Figure 2.1

Suggest an explanation for the fact that the variation shown in Figure 2.1 is linear.

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..... [2]

- (c) The thread in **2(b)** is now laid horizontally on a frictionless bench. A tensile force is applied to the ends of the thread so that there is a uniform stress throughout the thread of $6.8 \times 10^4 \text{ Pa}$.

- (i) Calculate the extension of the thread.

extension = m [2]

- (ii) Calculate the elastic potential energy of the thread.

elastic potential energy = J [2]

- (d) Suggest, with a reason, whether the total extension of the vertically suspended thread in **2(b)** is less than, equal to or greater than the answer in **2(c)(i)**.

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..... [1]

[Total: 12]